

Course on Probabilistic Design of Coastal Structures

Statistical Modelling of Environmental Conditions

04. CLIMATE EMULATOR

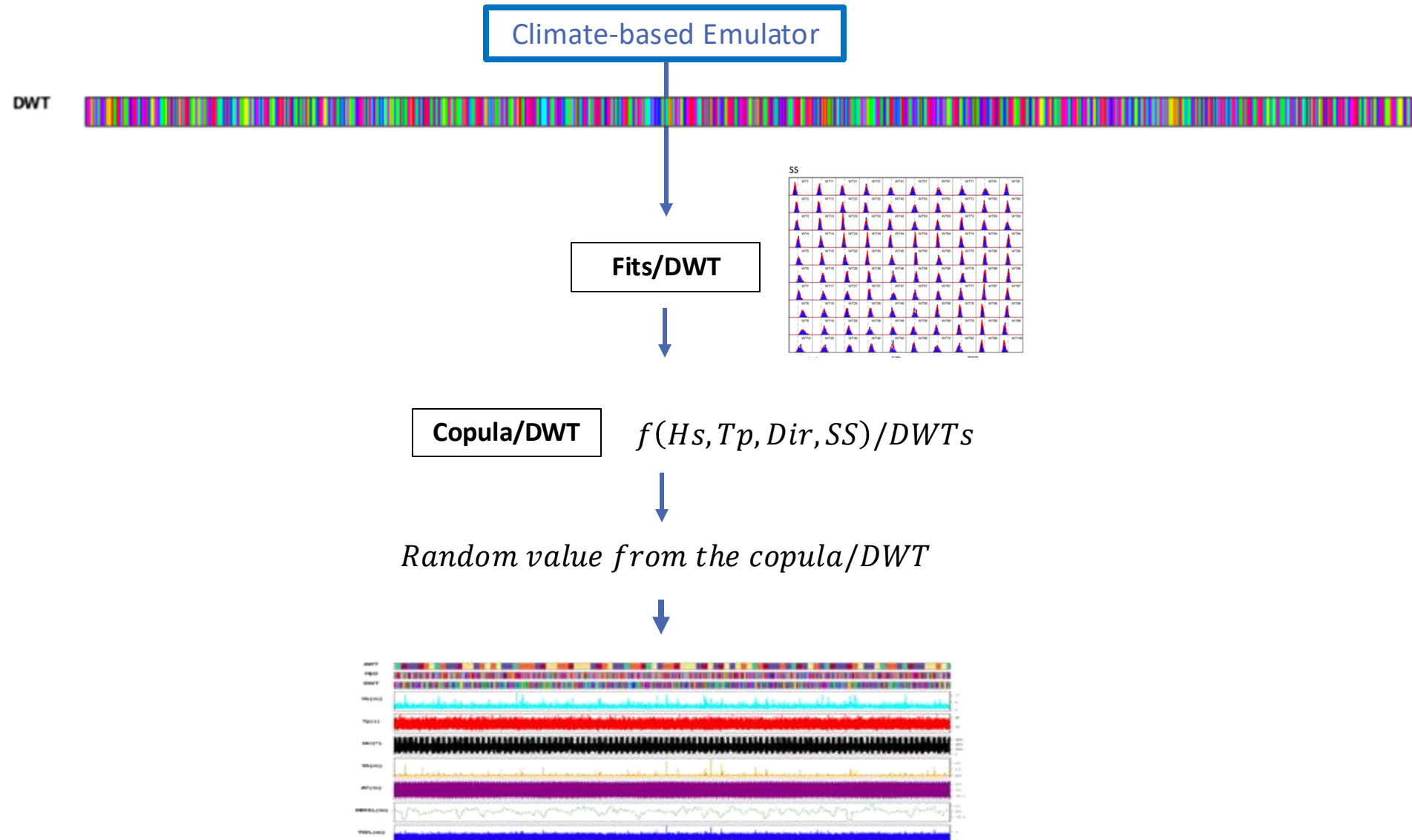
Jared Ortiz-Angulo Cantos

Researcher, GeoOcean Group, Universidad de Cantabria (UC)

ProCoast 2025
October 20th-23th, 2025



FRAMEWORK

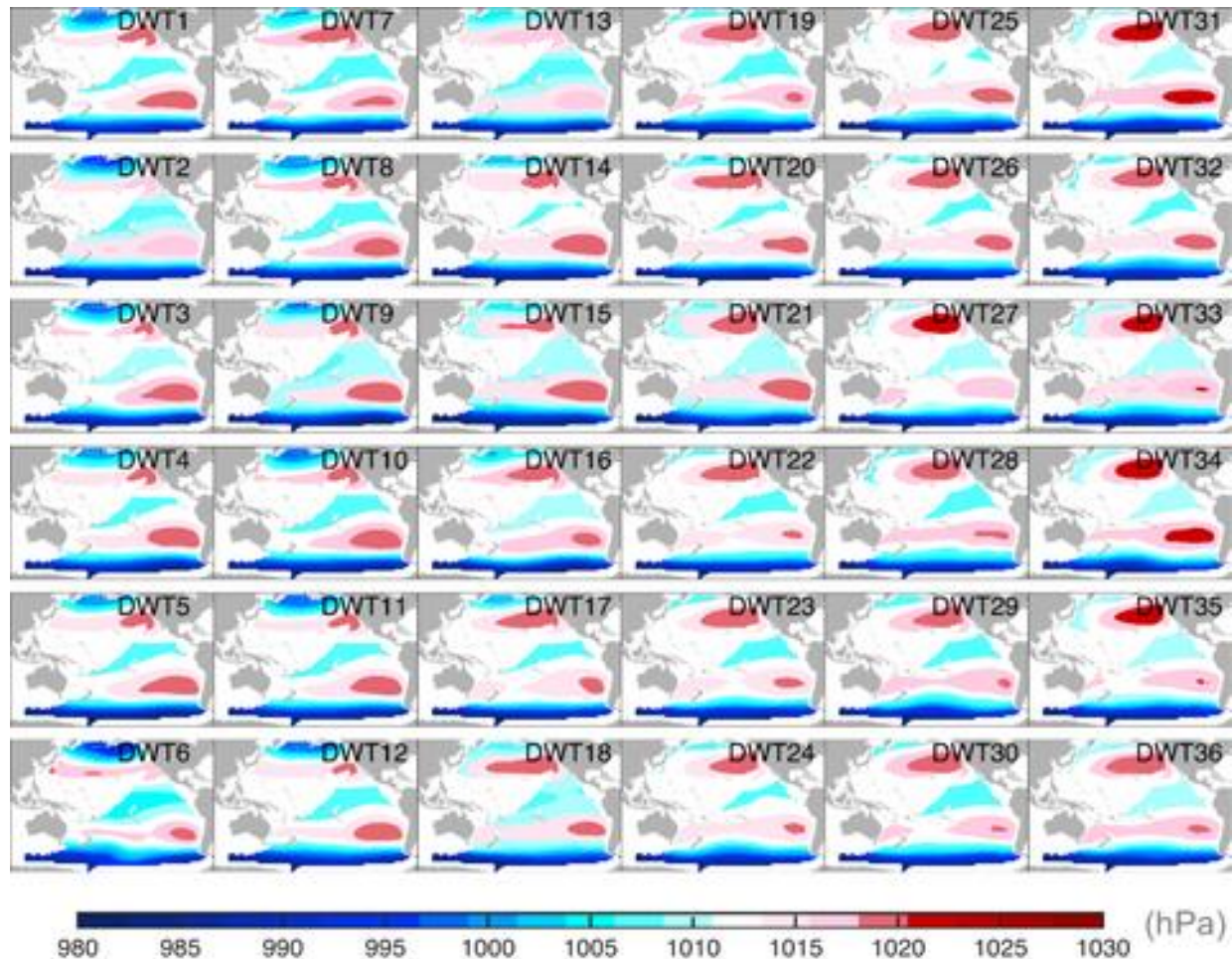


DISTRIBUTIONS

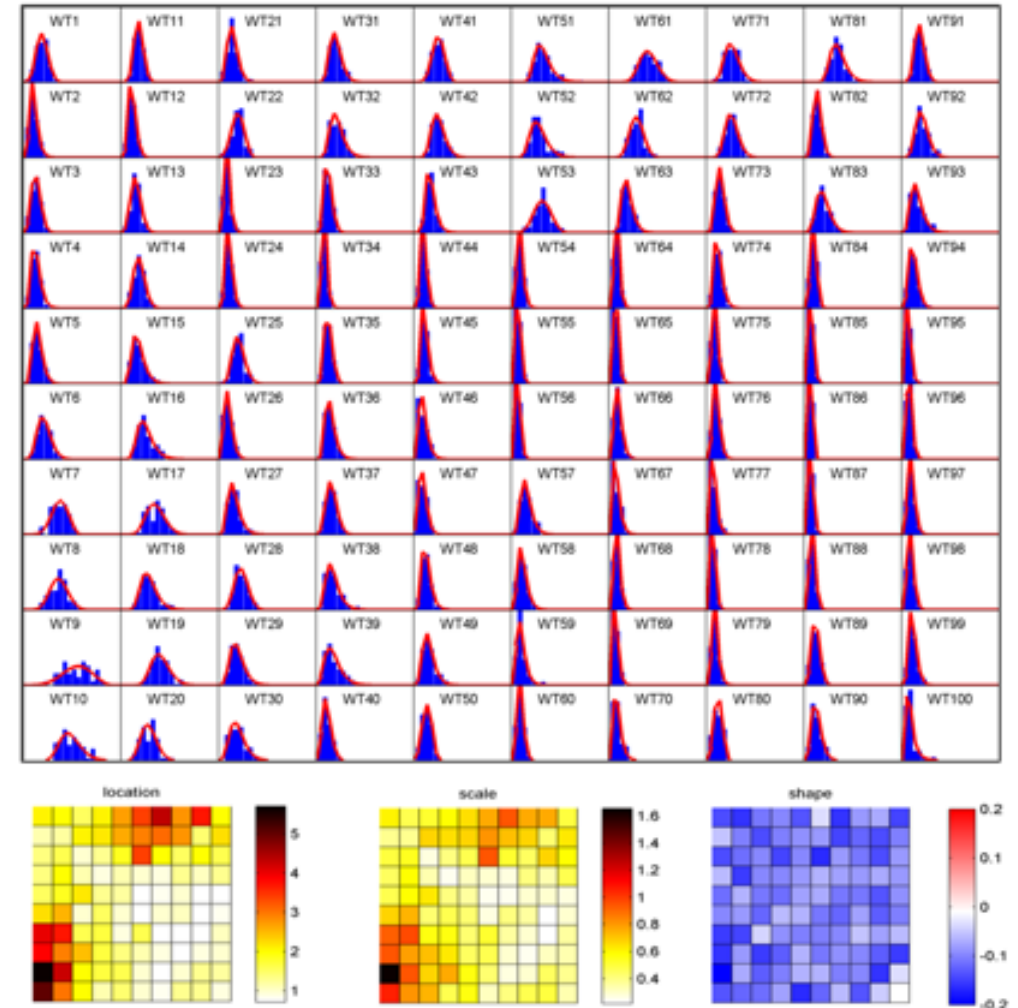
$$F(y) = \exp \left\{ - \left[1 + \xi \left(\frac{y - \mu}{\psi} \right) \right]^{\frac{-1}{\xi}} \right\};$$

Marginal distributions associated to each WT

Rueda et al, 2016



Hs

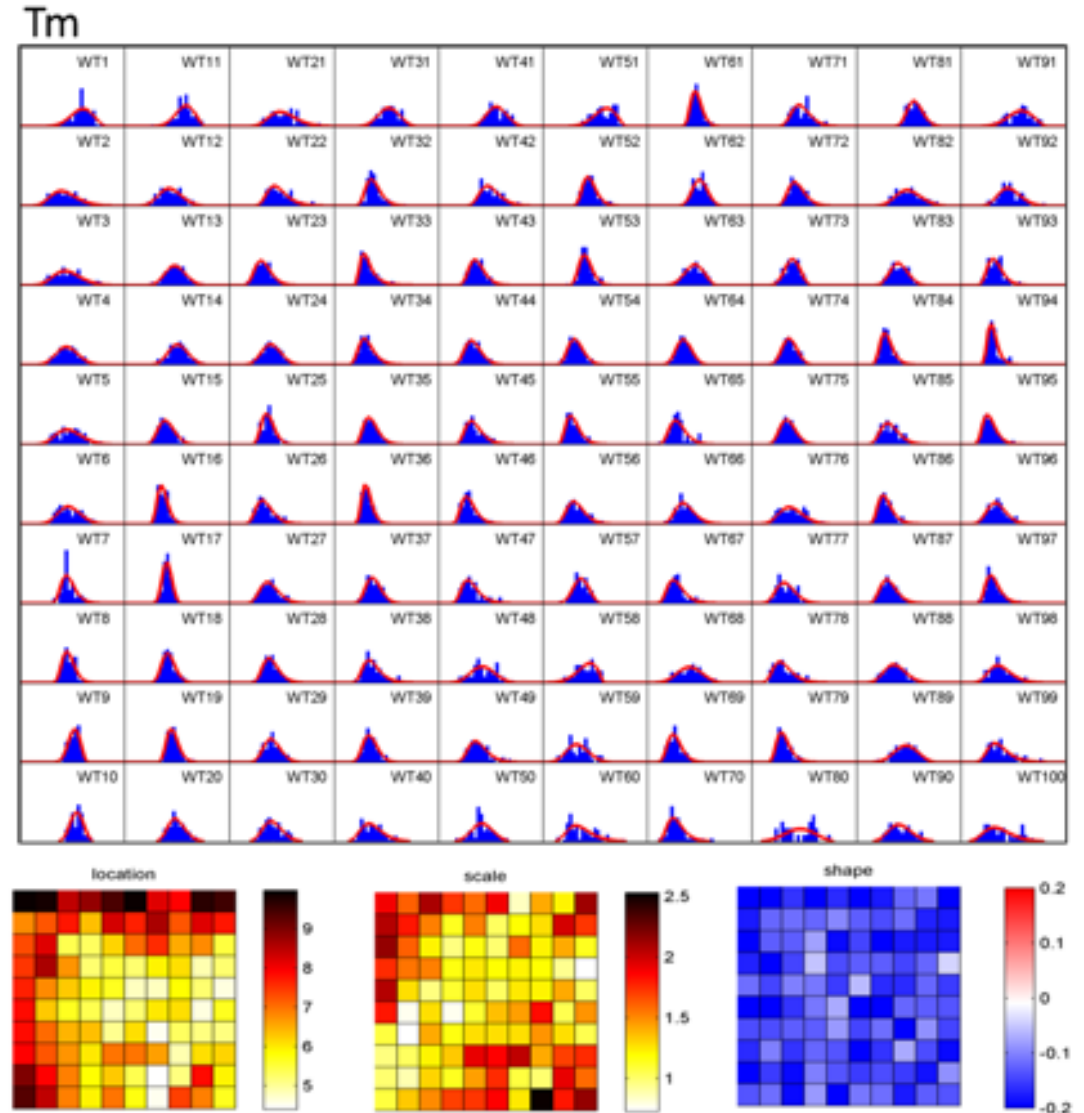
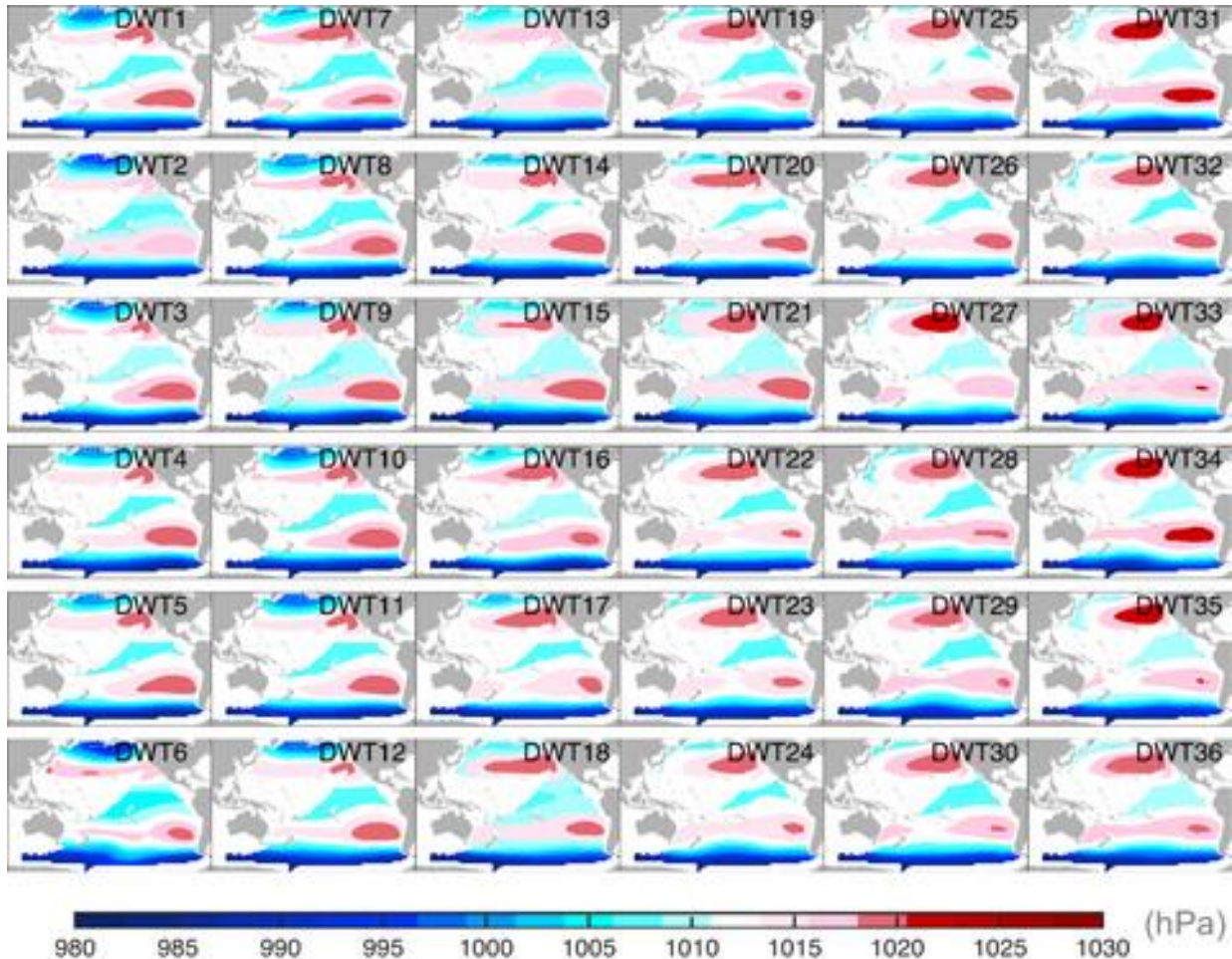


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Rueda et al, 2016

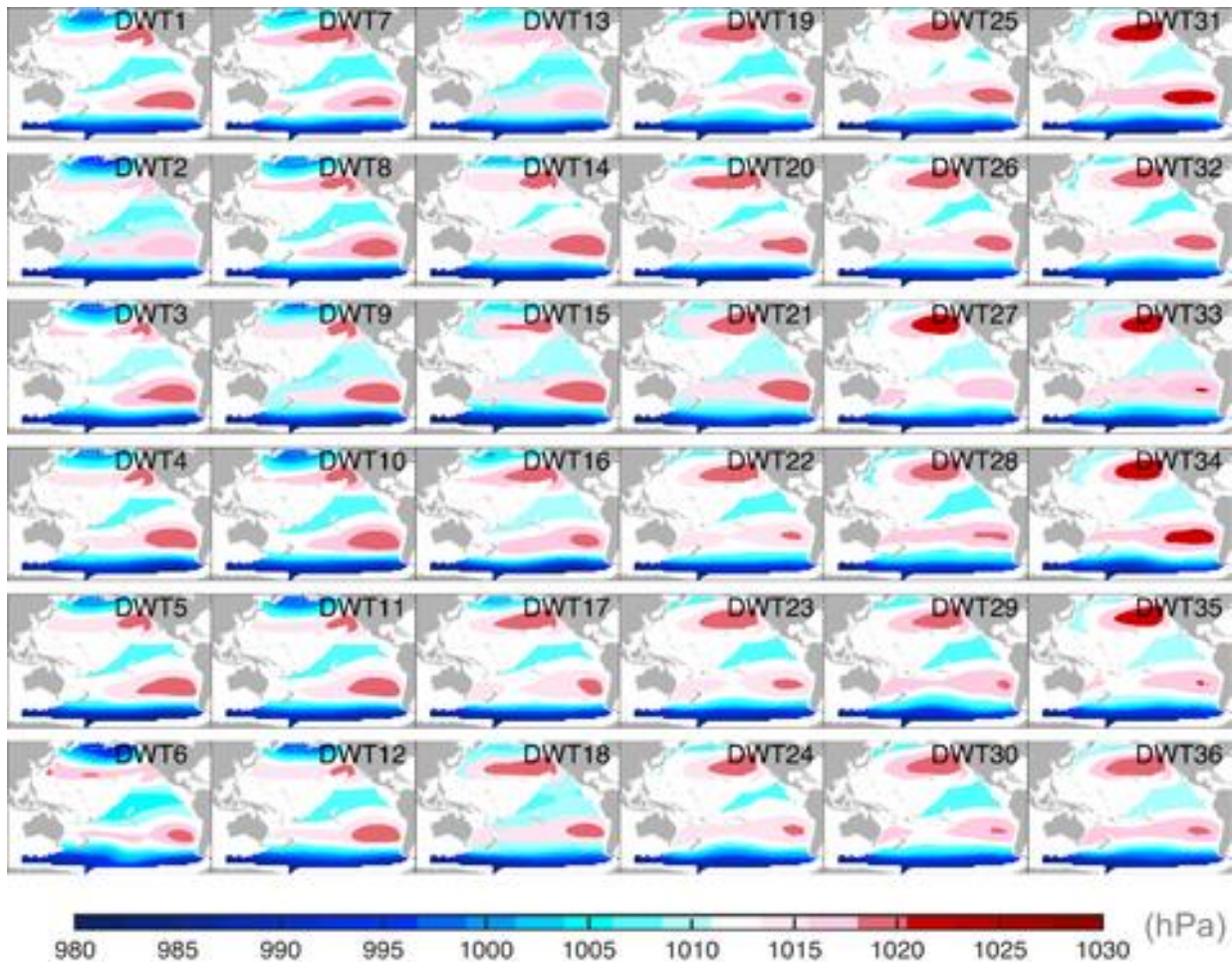


DISTRIBUTIONS

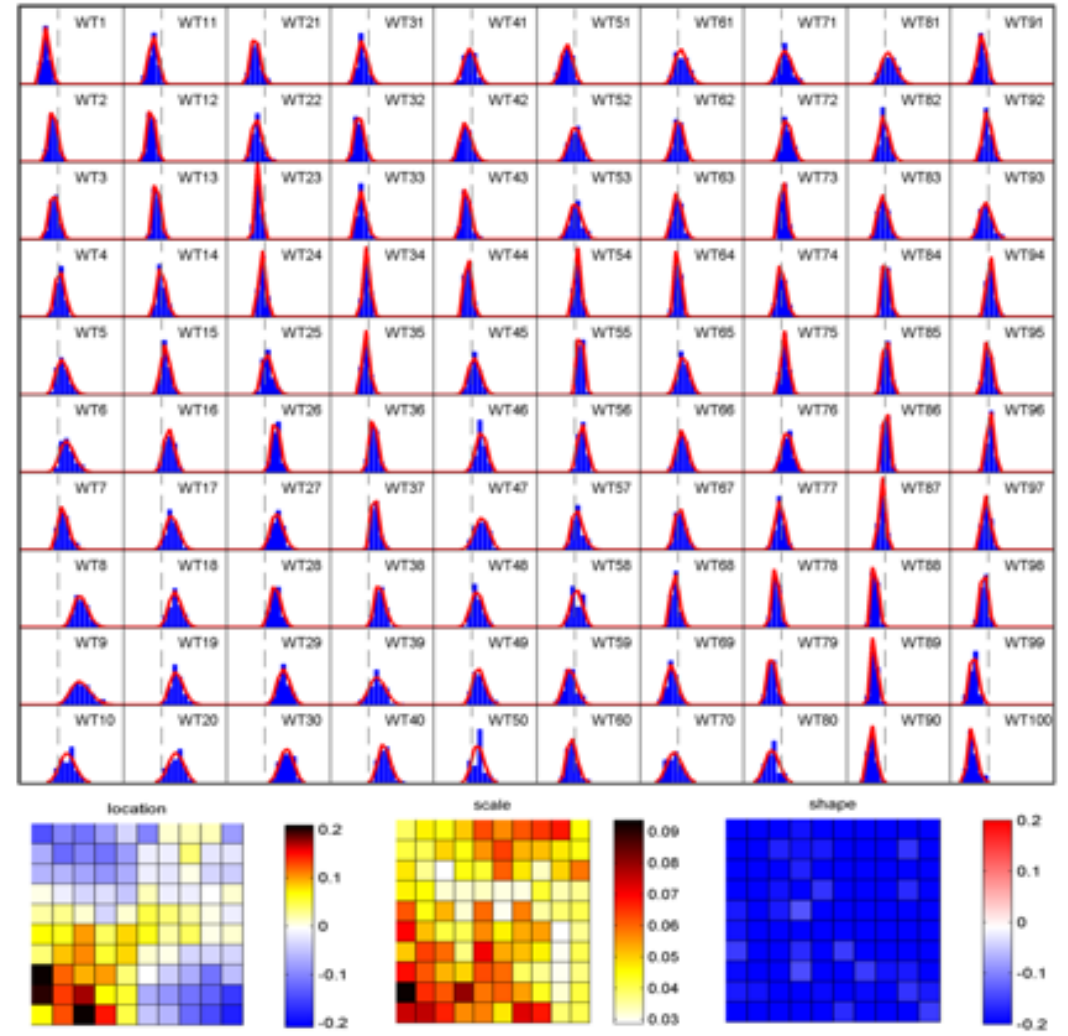
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Rueda et al, 2016

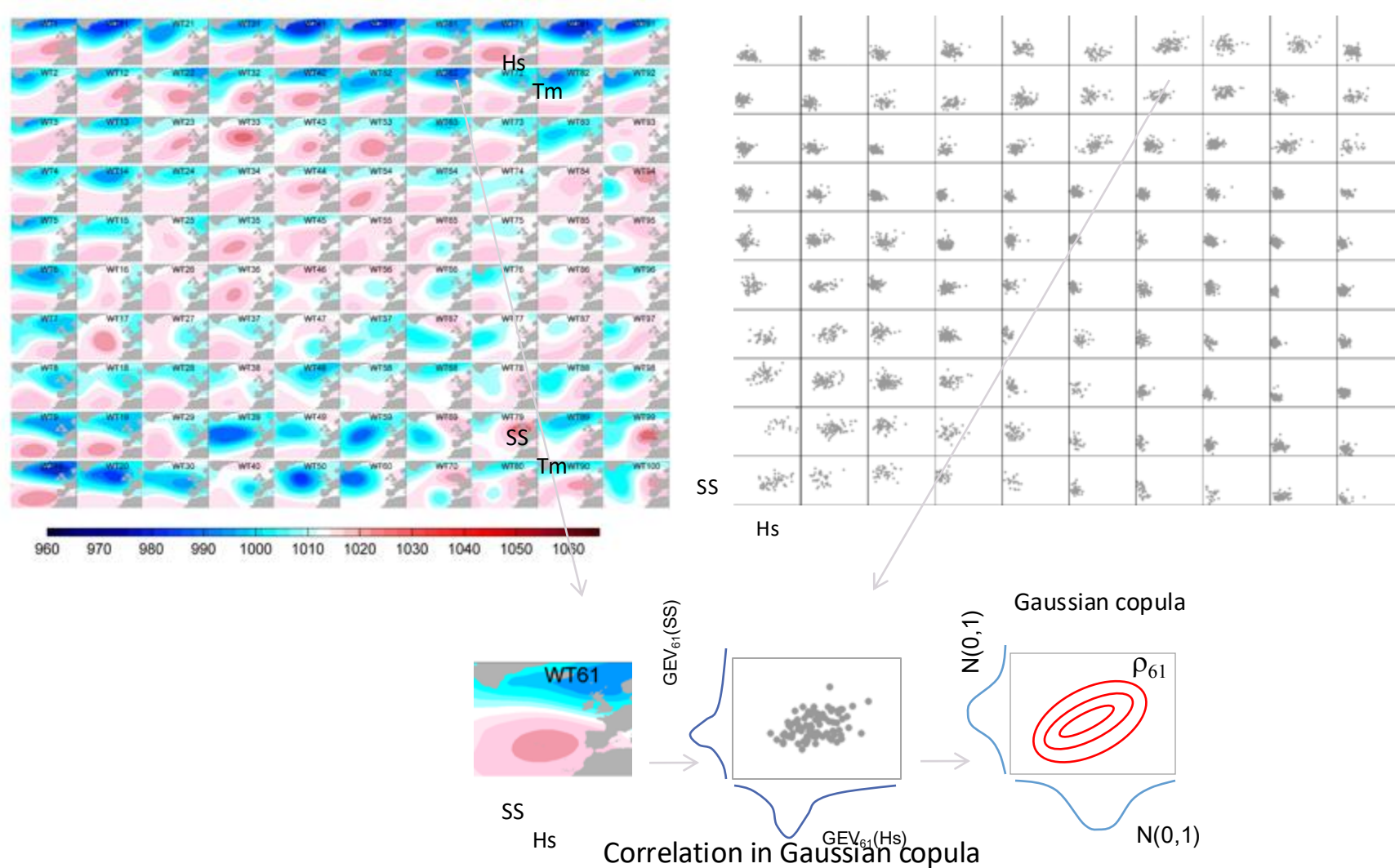


SS



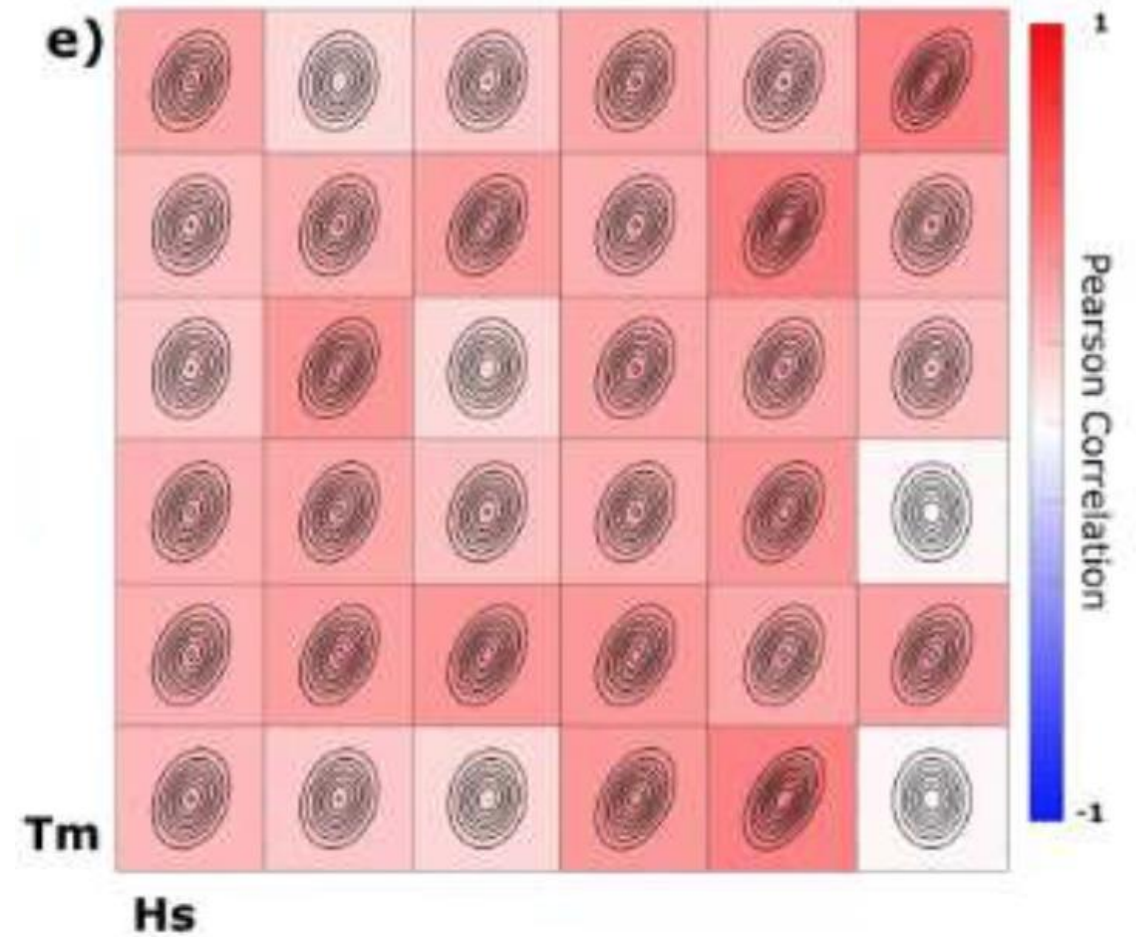
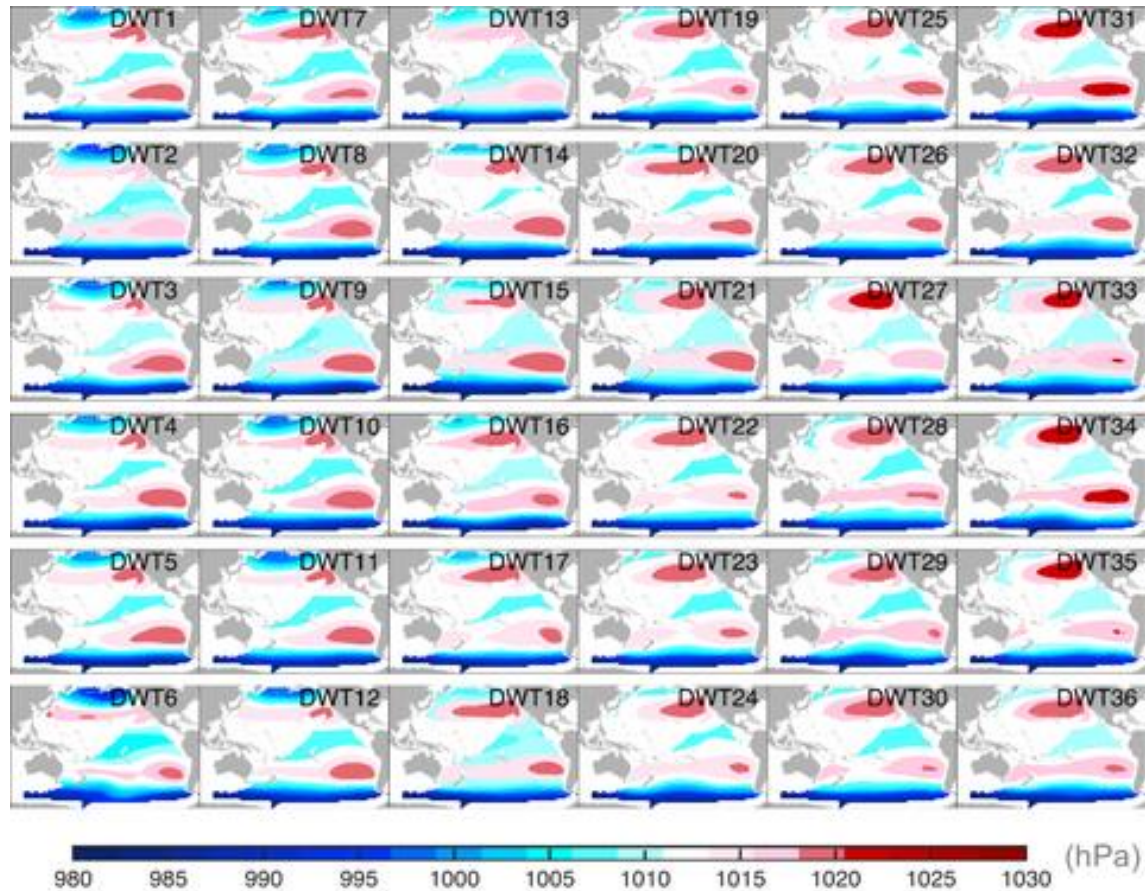
COPULAS

Dependence structure



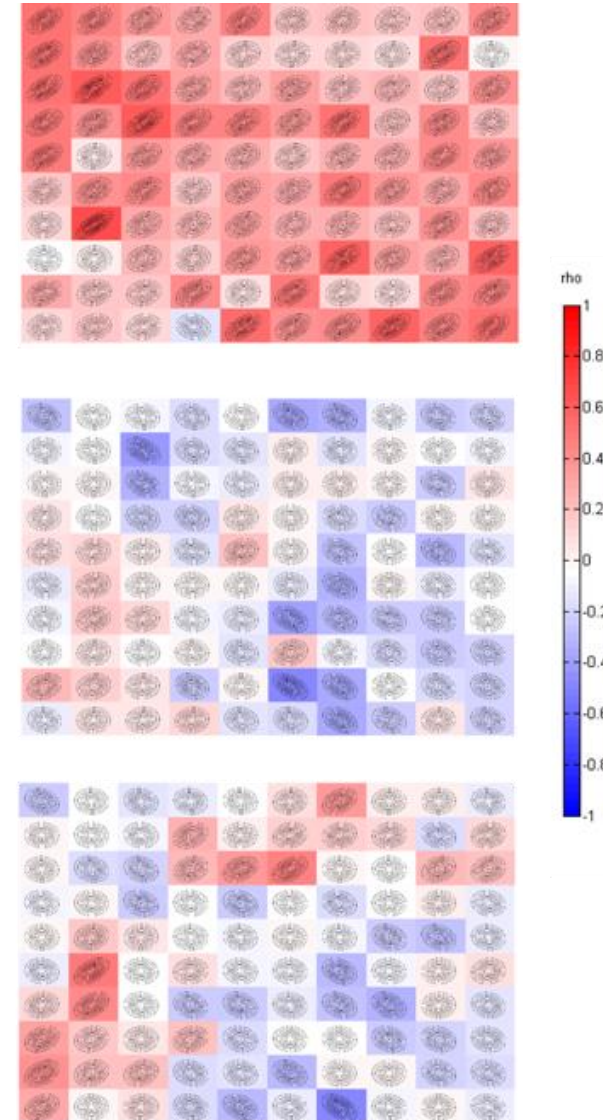
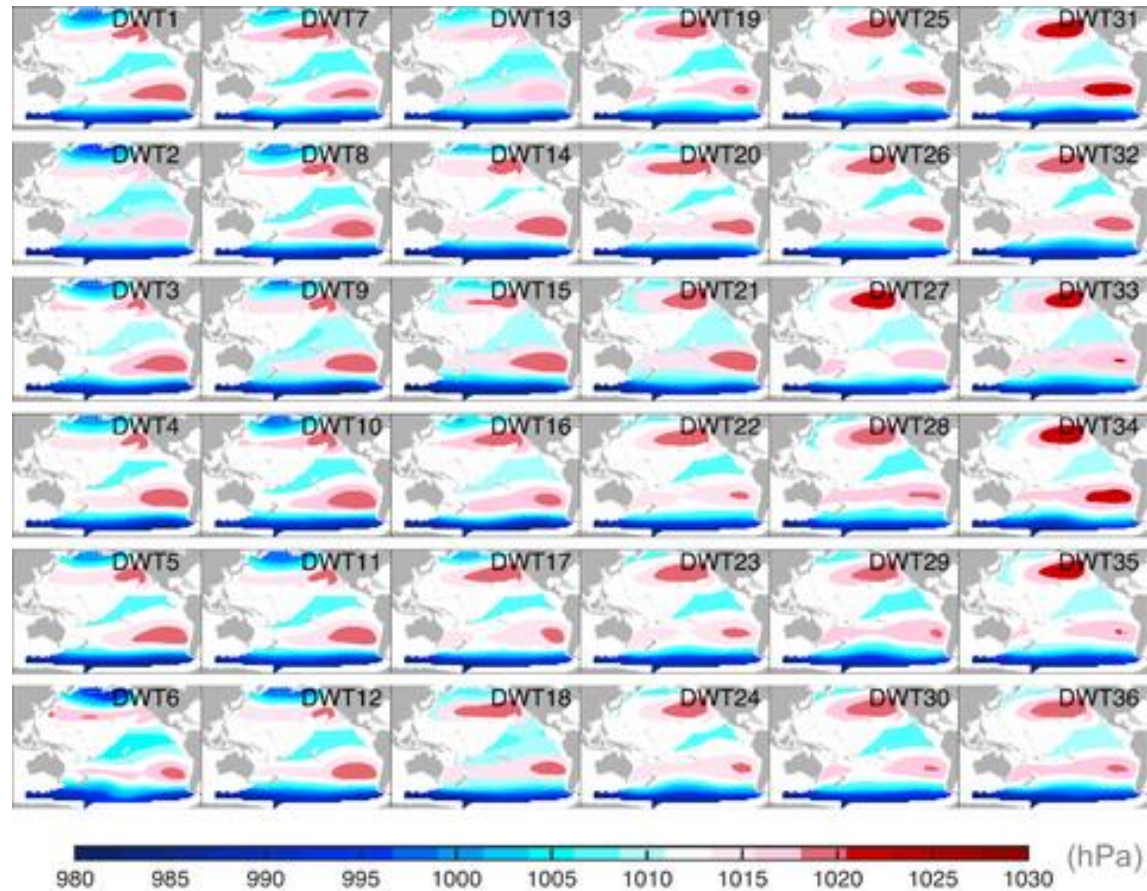
COPULAS

Rueda et al, 2016



COPULAS

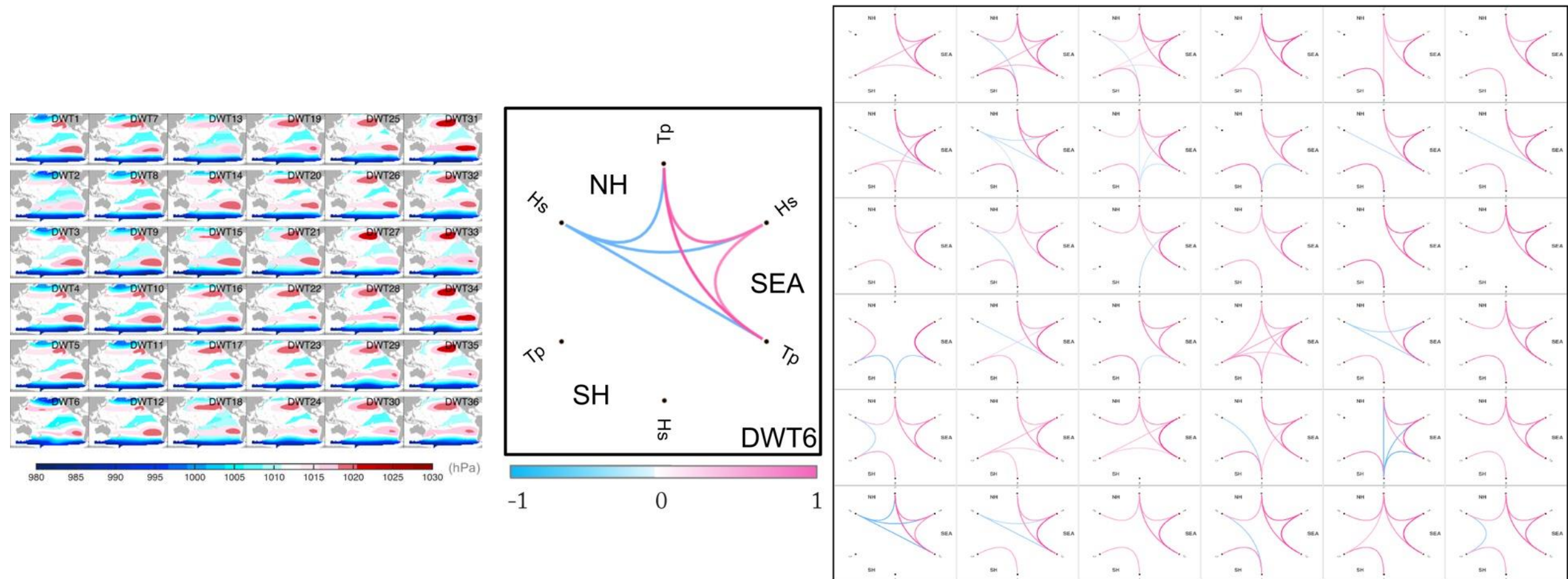
Rueda et al, 2016



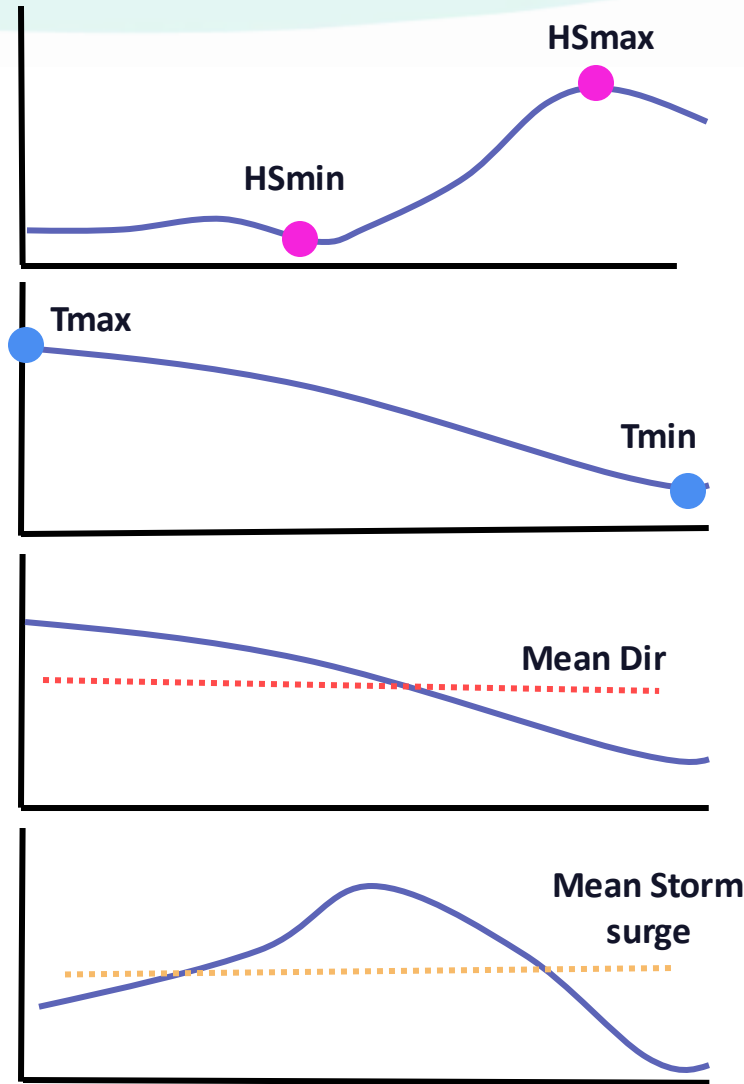
COPULAS

Rueda et al, 2016

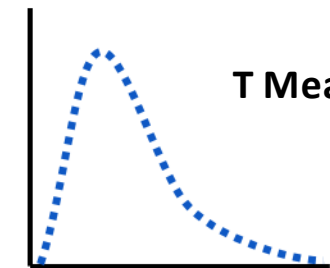
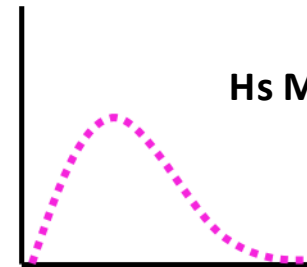
Copulas are able to emulate realistic combinations of variables based on the historical probabilities.



COPULAS



GAUSSIAN COPULA



WT9

WT 1

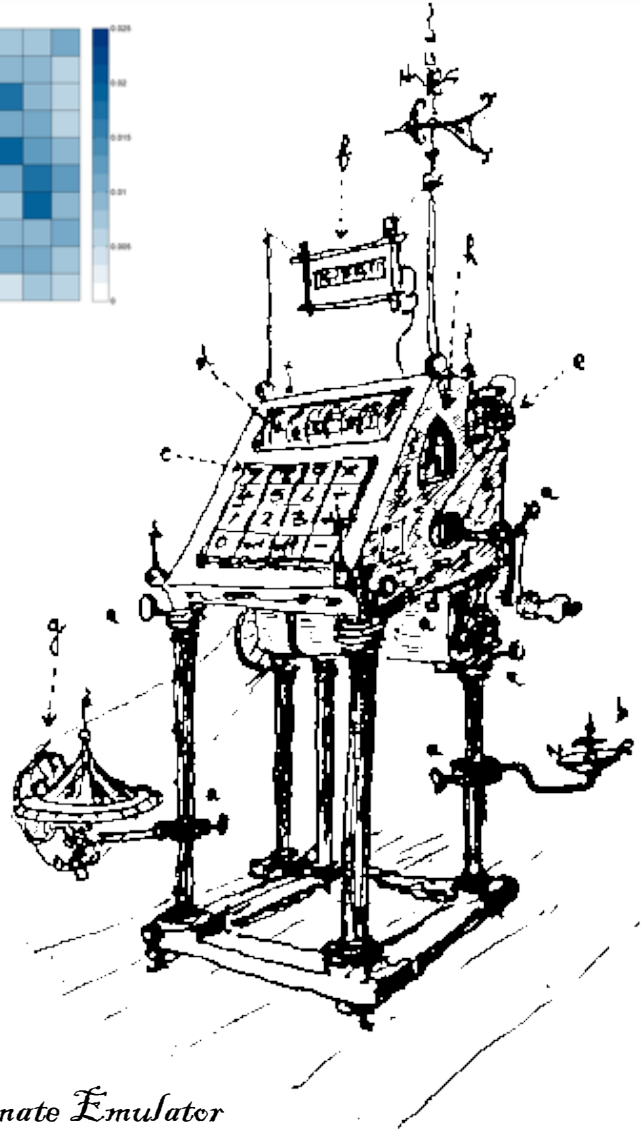
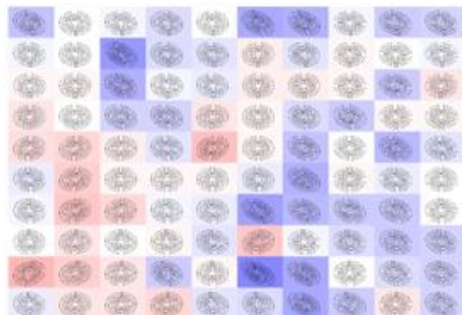
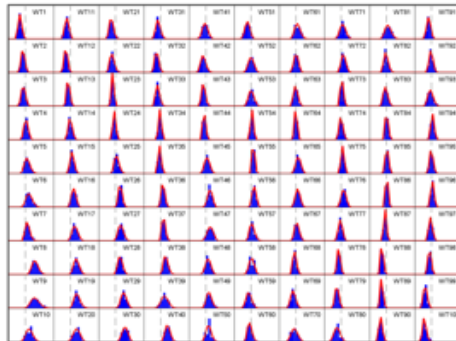
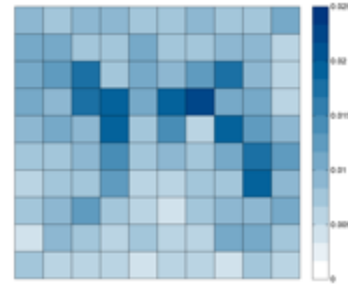
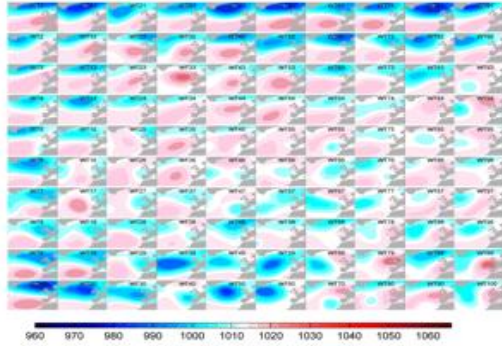
WT1

WT1

WT1

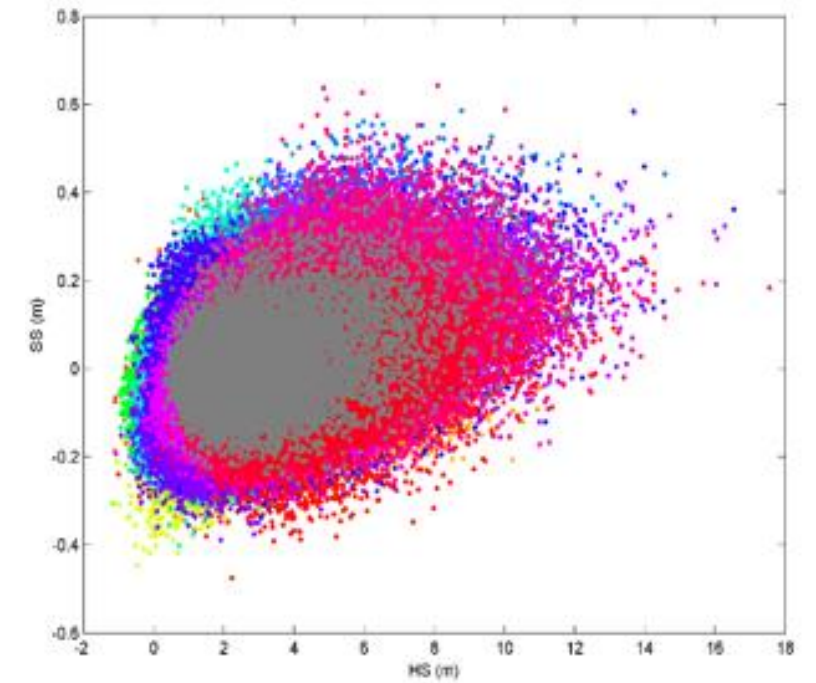
WT15

EMULATOR



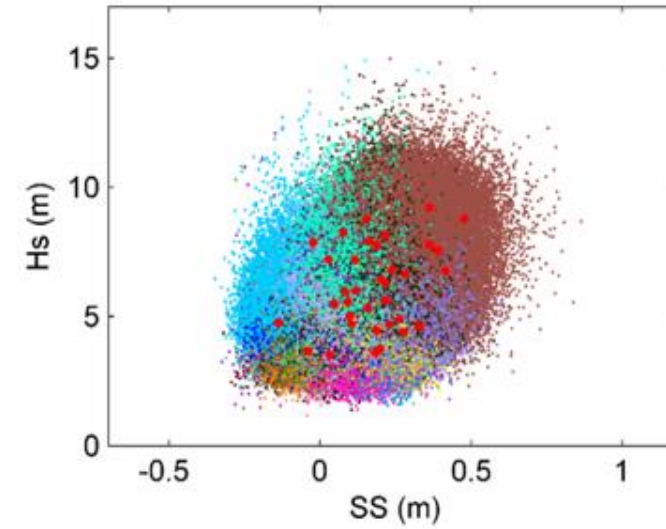
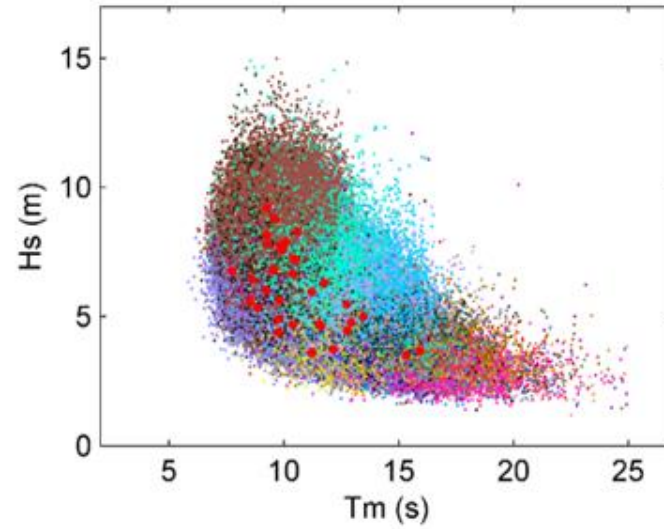
Climate Emulator

Multivariate simulation



EMULATOR

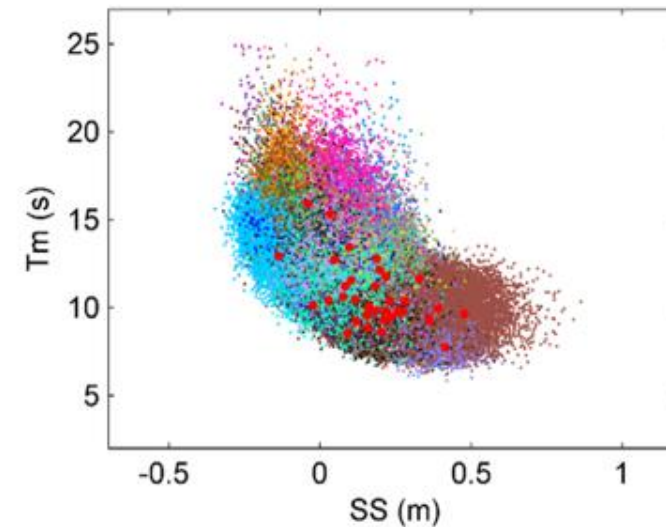
Multivariate simulation



WT of origin



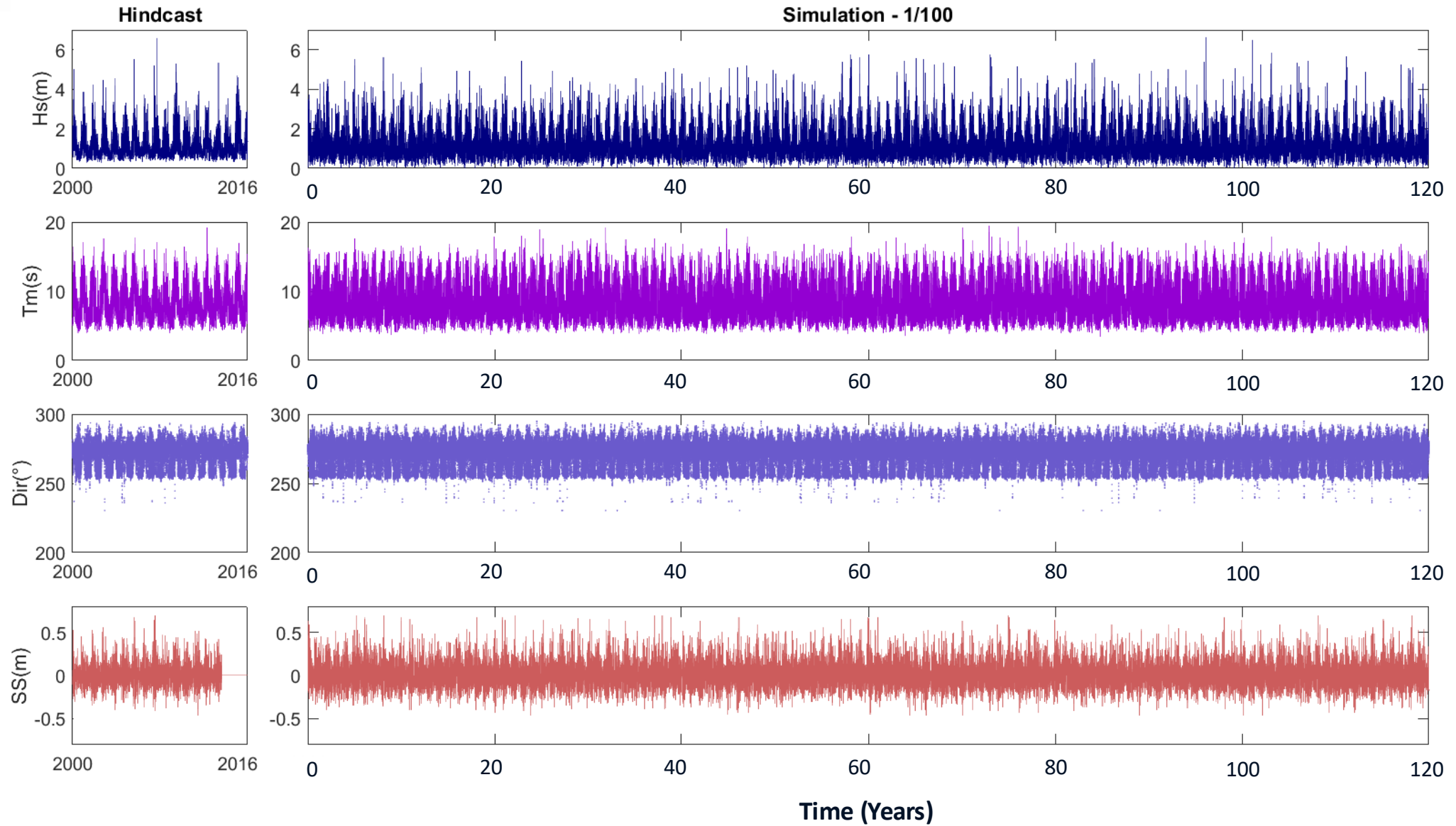
450.000 simulated years



• Historical data

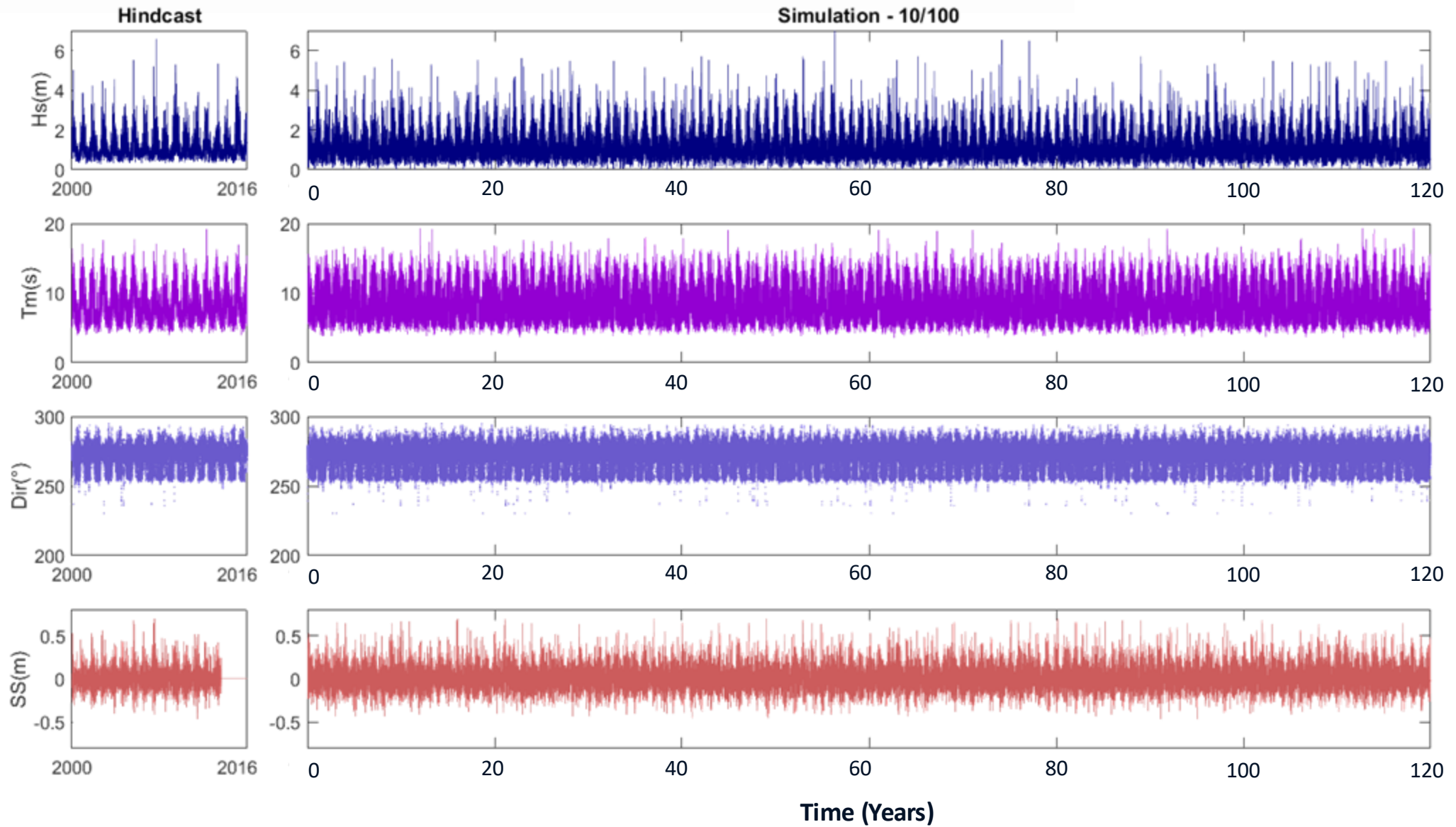
RESULTS

Imperial Beach



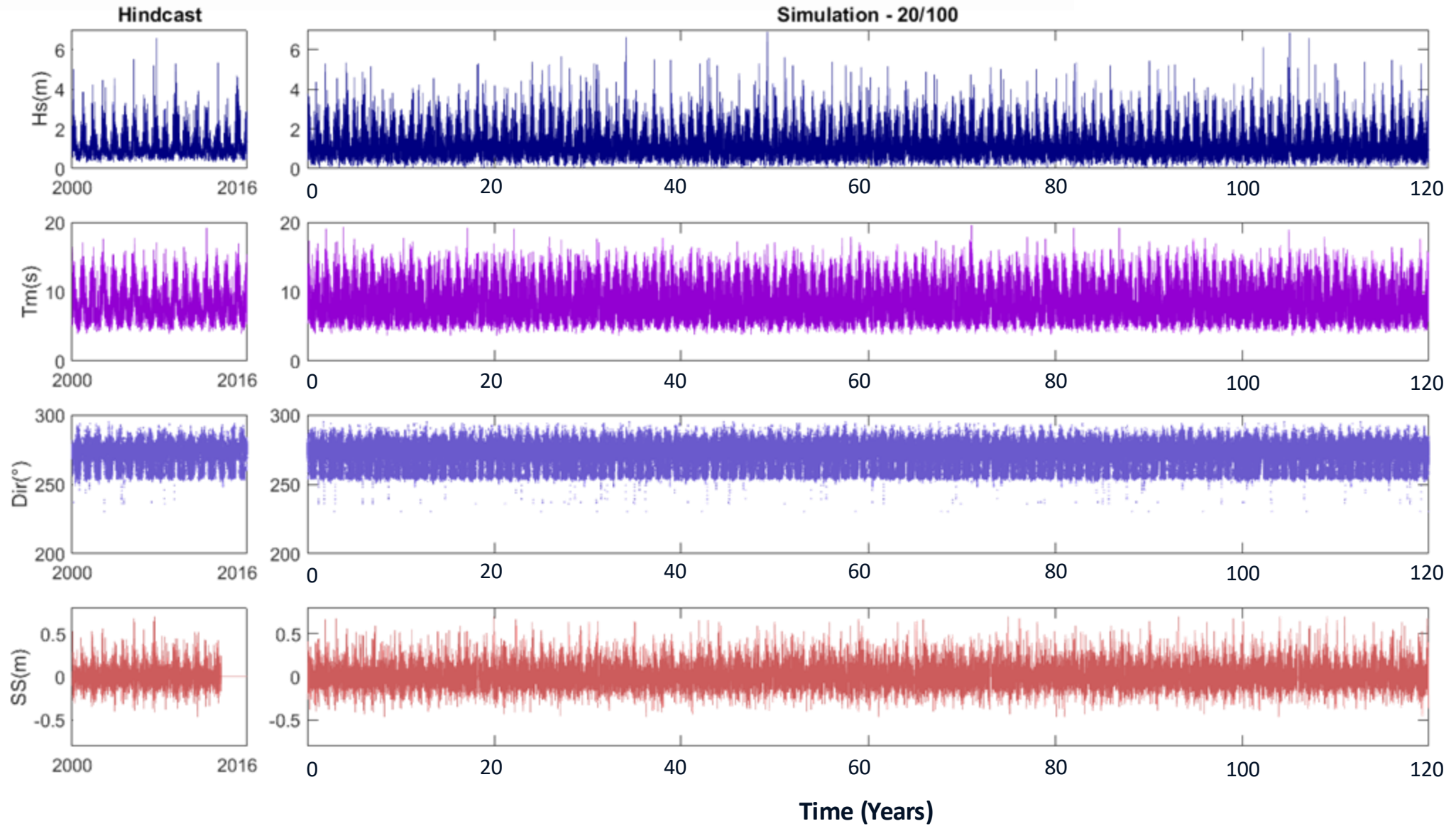
RESULTS

Imperial Beach



RESULTS

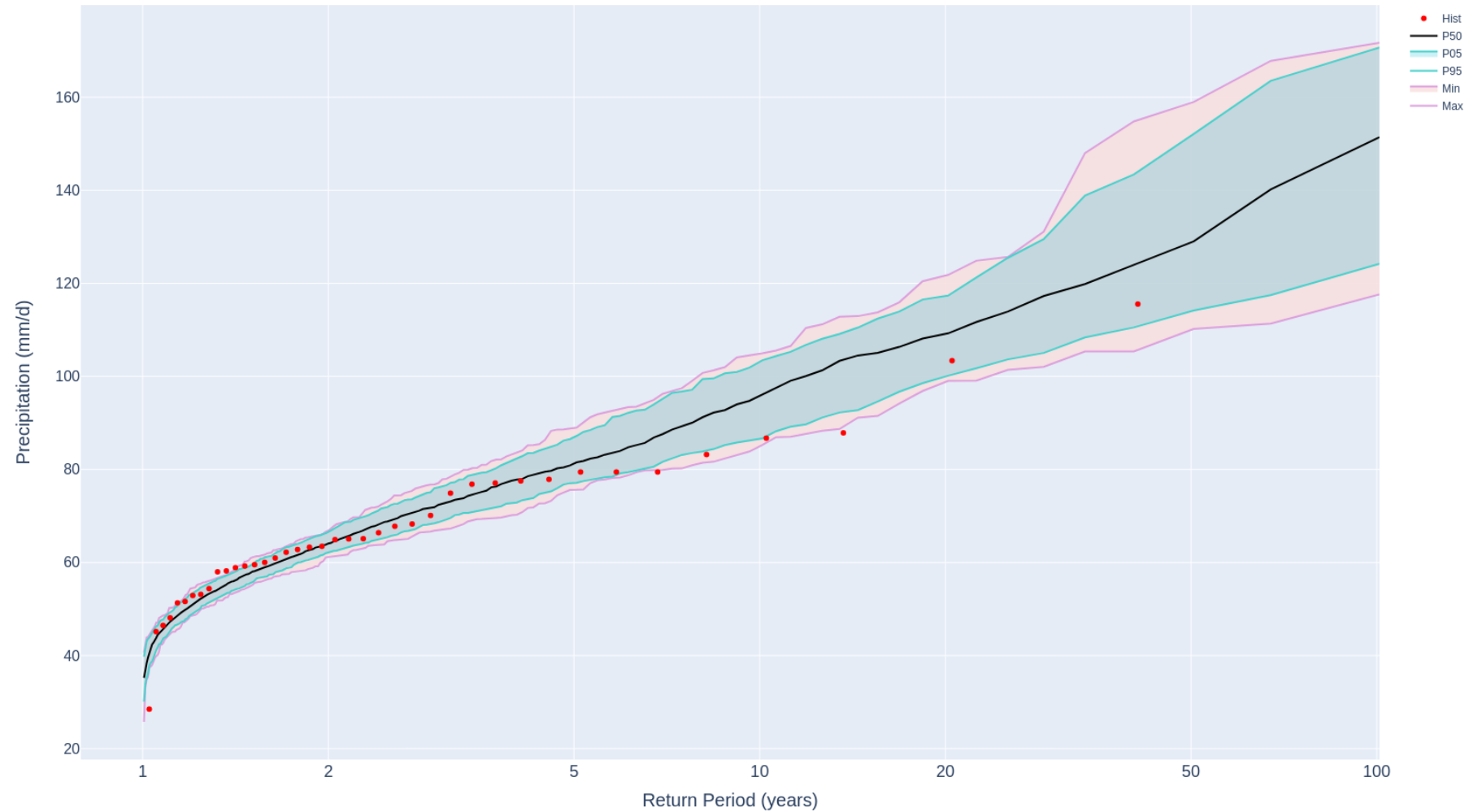
Imperial Beach



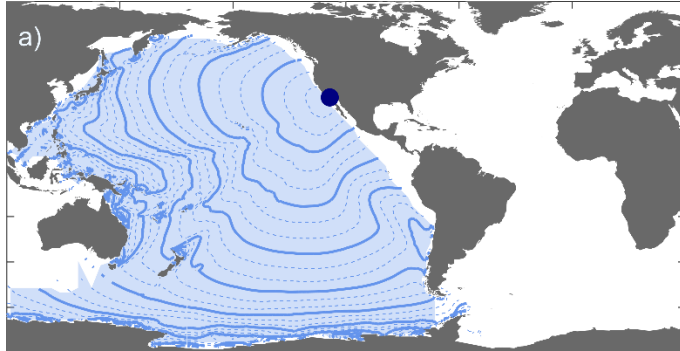
RESULTS

RETURN PERIOD OF THE ANNUAL MAXIMUMS

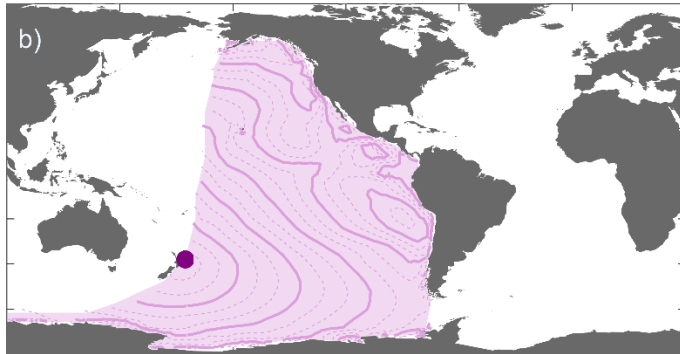
100 Simulations x 100 years



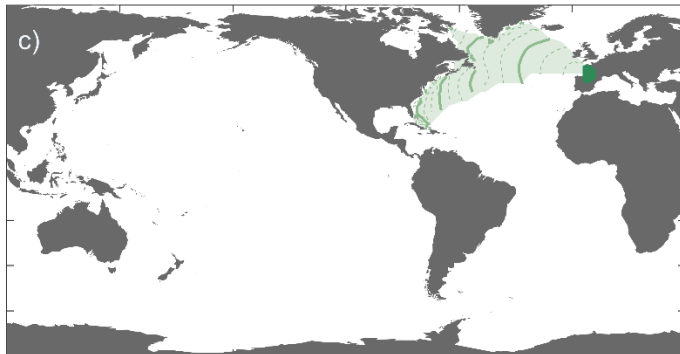
Applications



LONG-SHORE WAVE POWER : IMPERIAL BEACH



CROSS-SHORE EROSION : TAIRUA BEACH

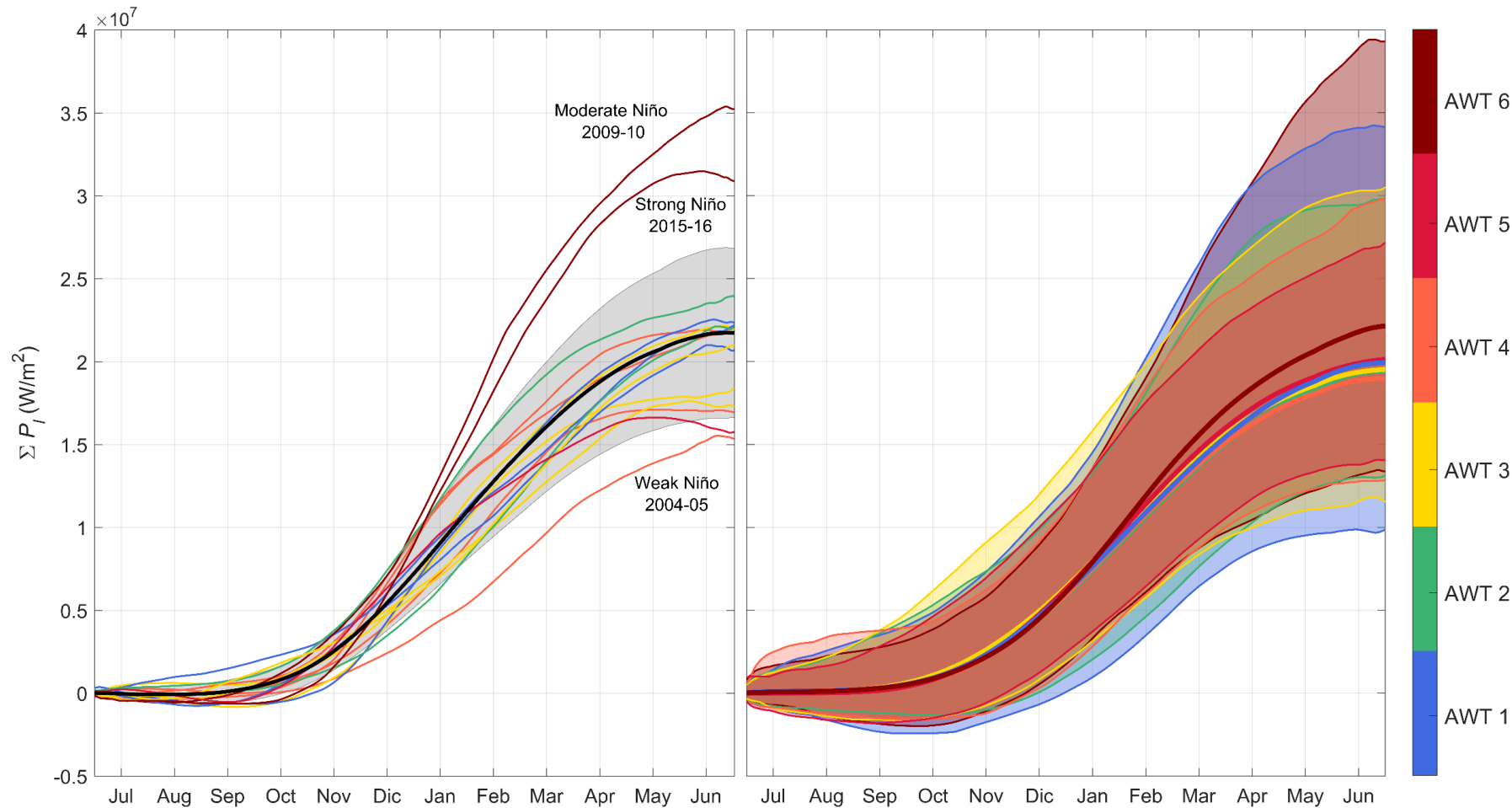


BEACH ROTATION : LAREDO

Applications

CERC FORMULA (Komar, 1998)

$$P = ECn \sin(\theta) \cos(\theta) = \frac{1}{8} \rho_w g H^2 Cn \sin(\theta) \cos(\theta)$$



6° north shift on
directions during El
Niño 2009-2010
(Barnard et al., 2011)

Imperial beach highly
eroded and flooded
2015-2016
(Young et al., 2018)

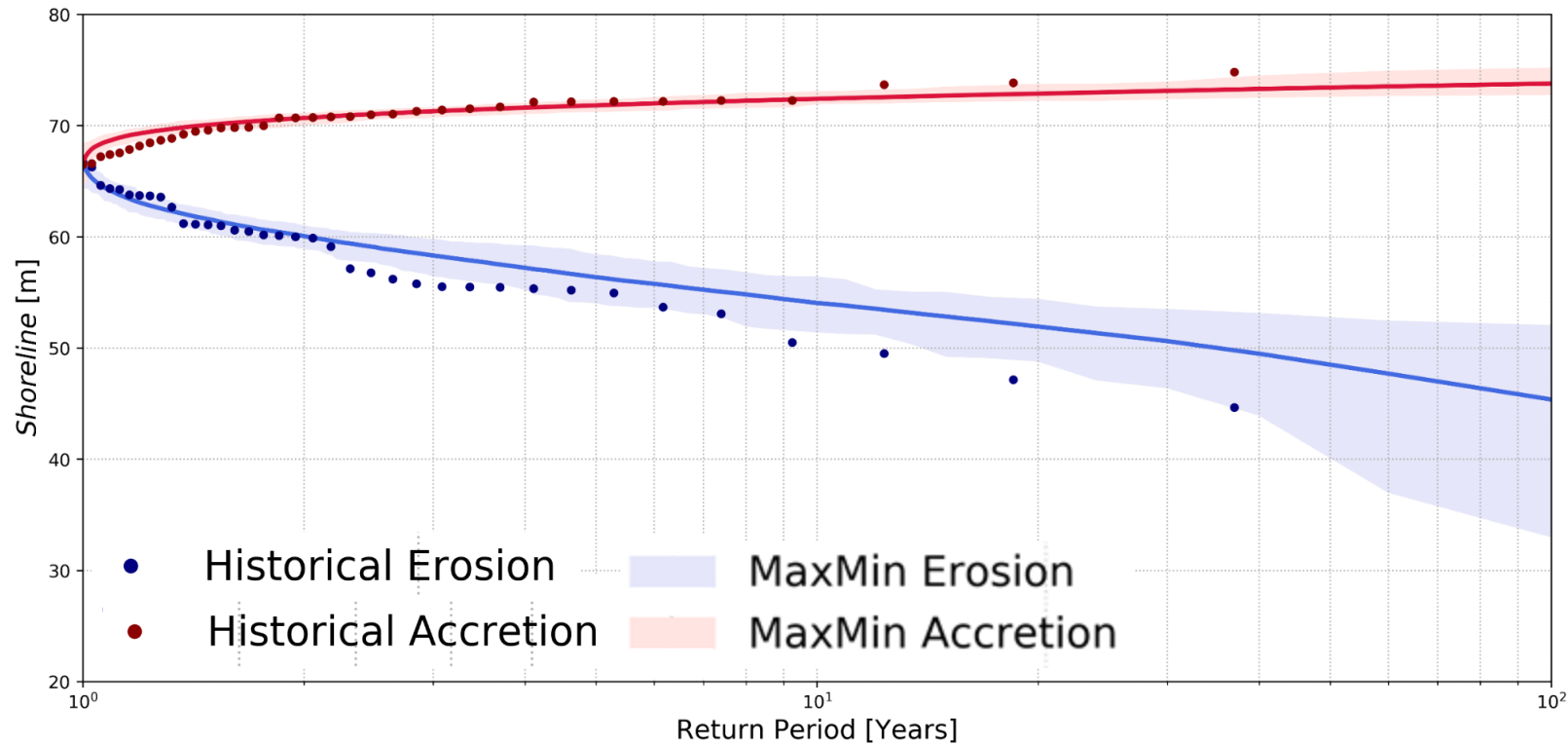
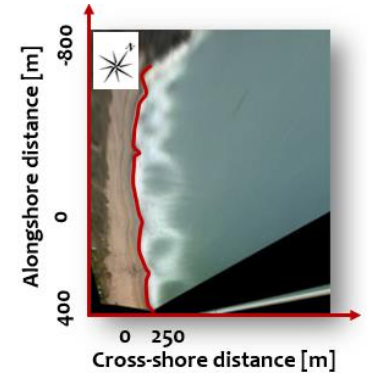
More southerly waves
during La Niña

Applications

Yates et al., 2009
Cross-Shore equilibrium model

$$\frac{dy}{dt} = C^{\pm} E^{1/2} (E - E_{eq})$$

18 years of daily
shoreline position from
a camera system

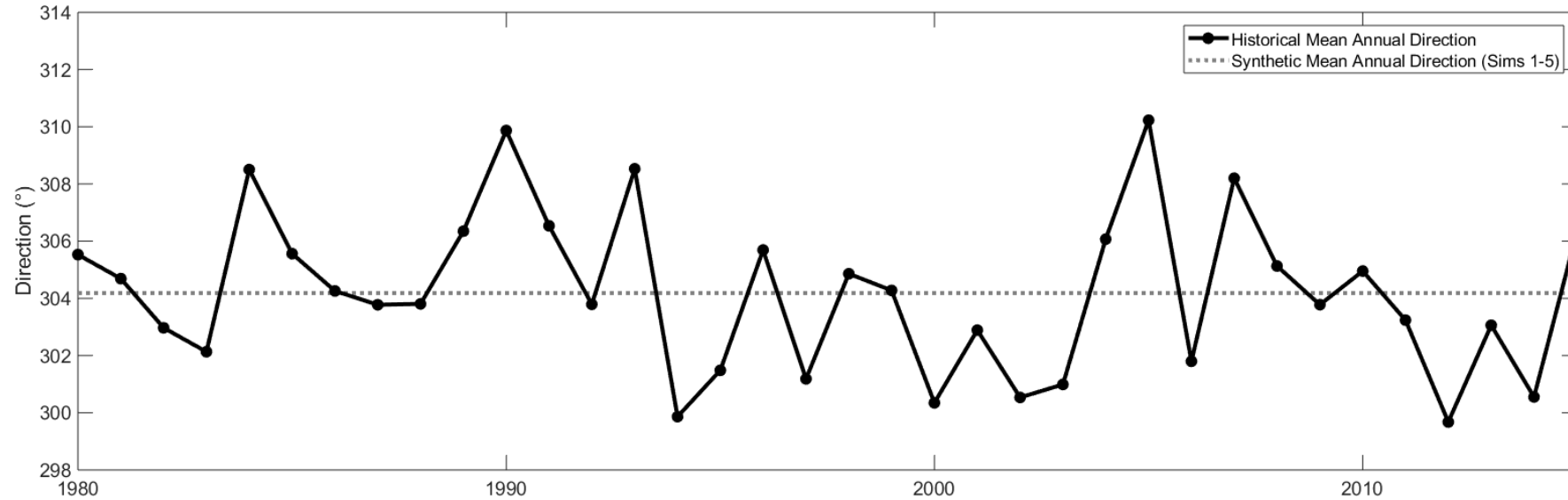


Applications

Shoreline change at seasonal
to interannual scales

Triggered by changes on the
mean wave direction

Annual mean wave direction
proxy

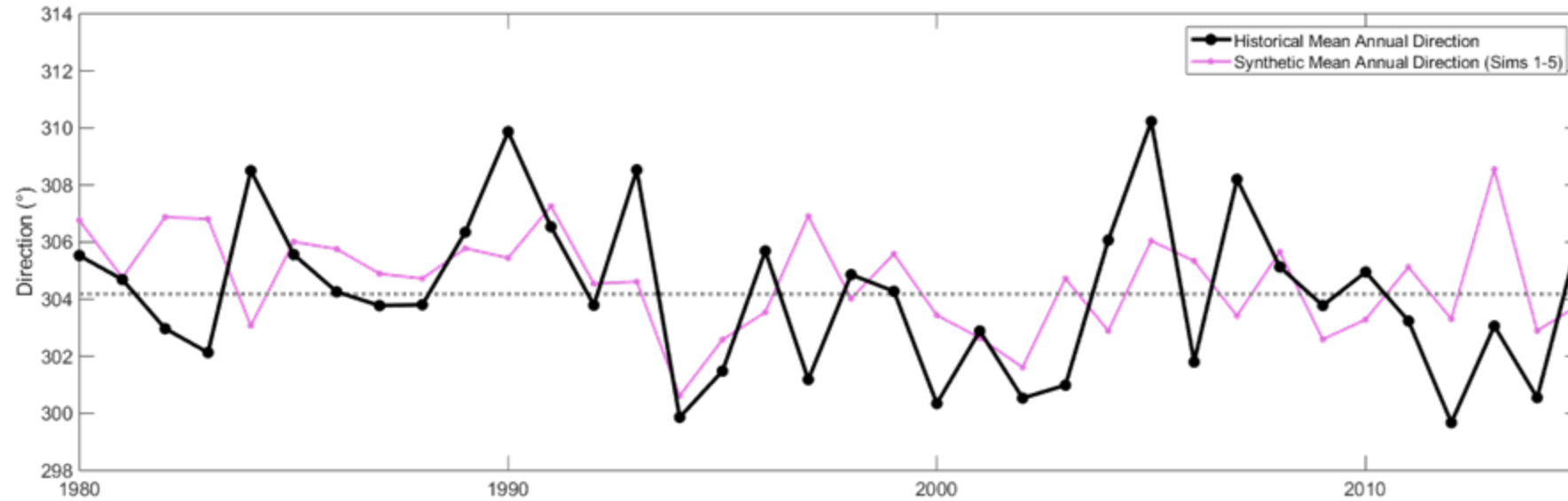


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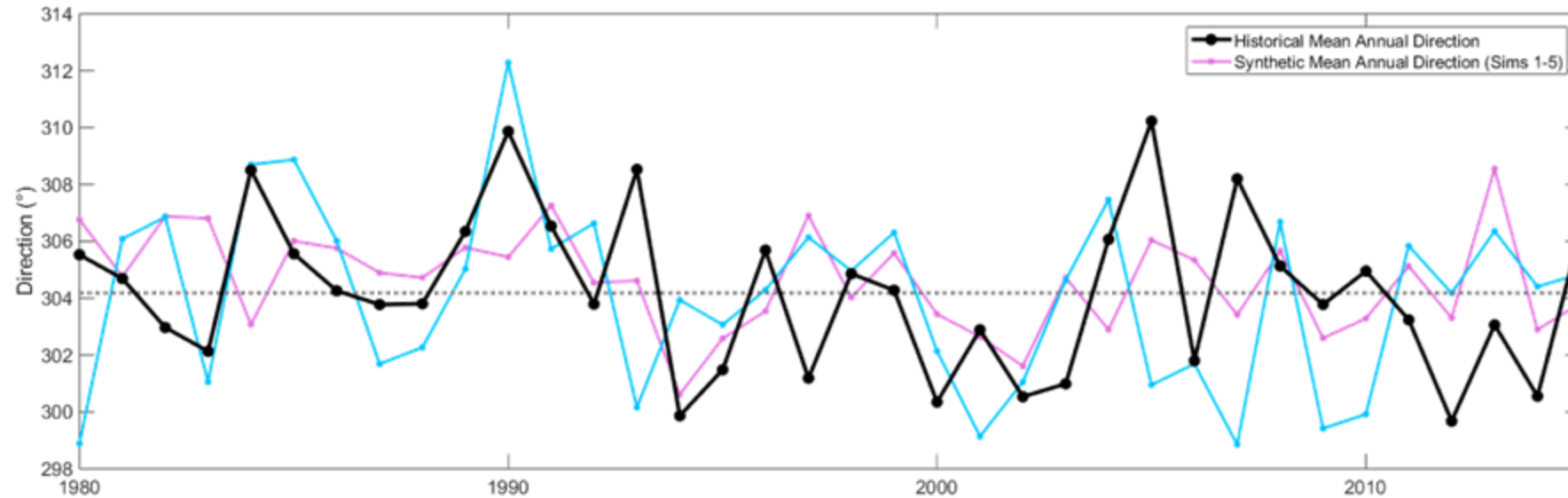


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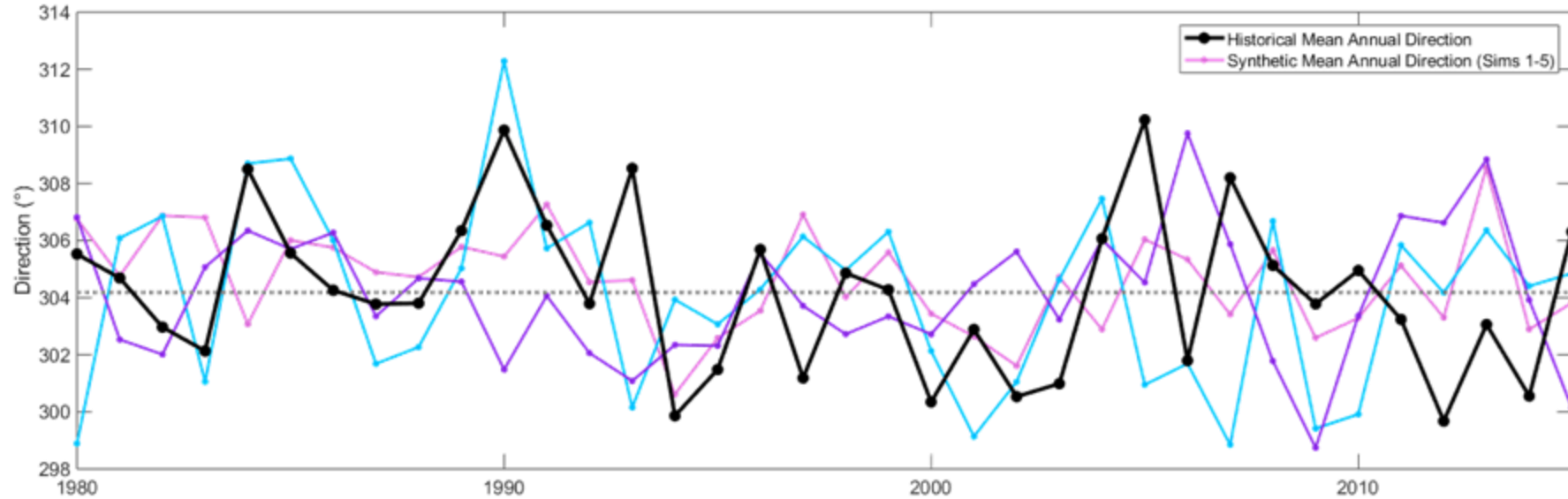


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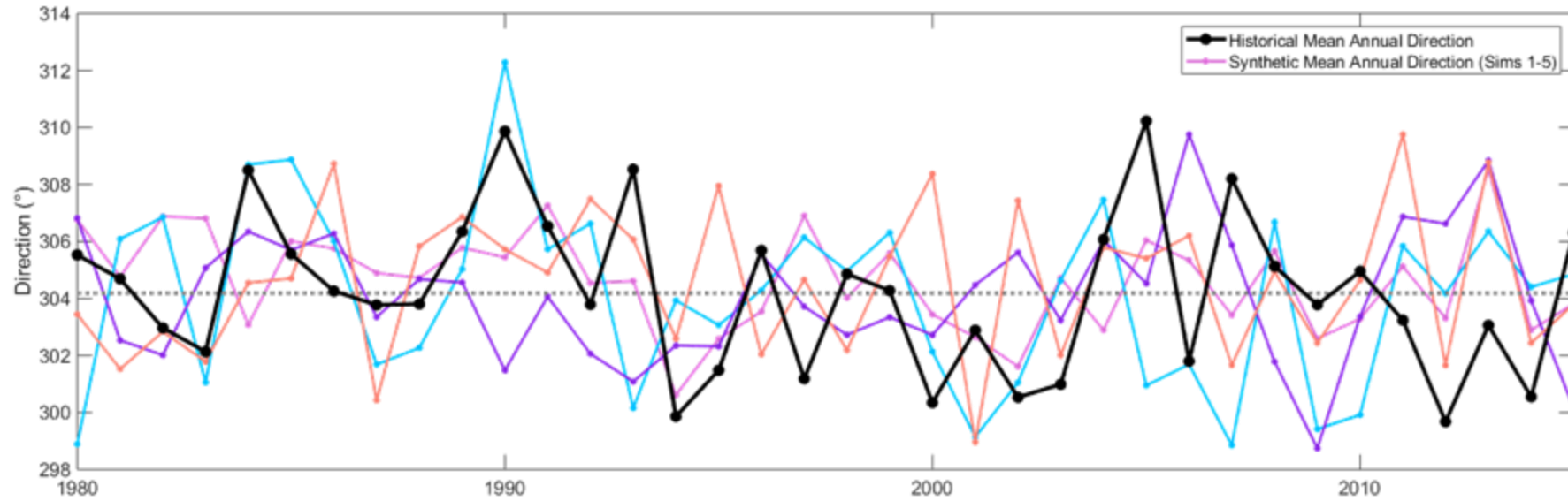


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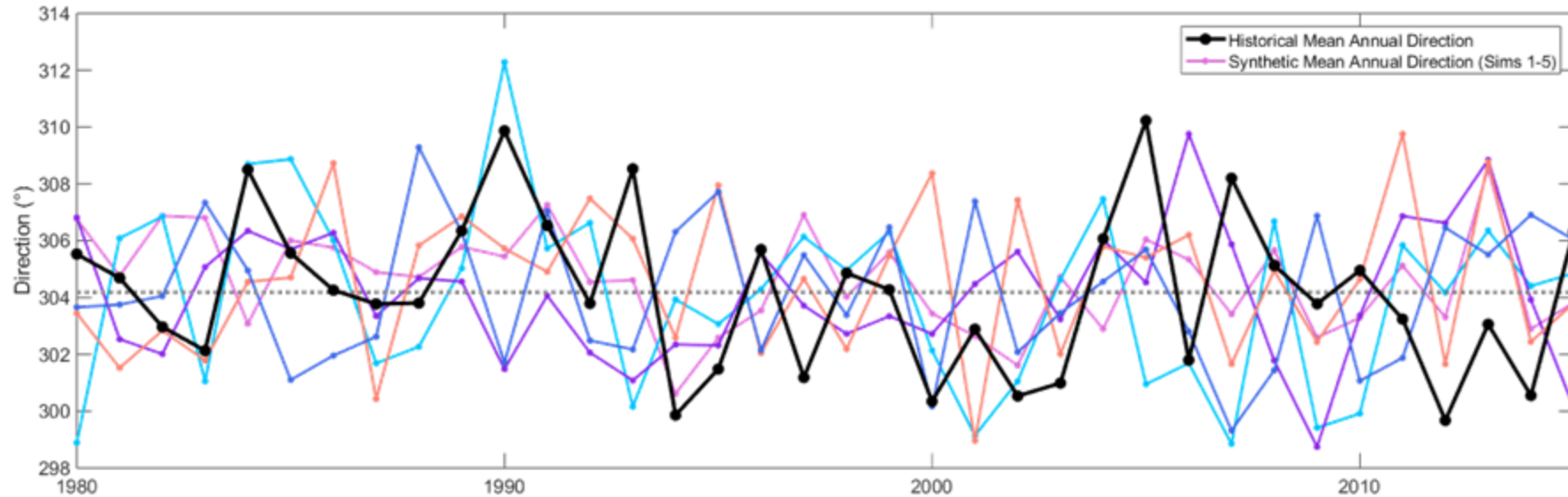


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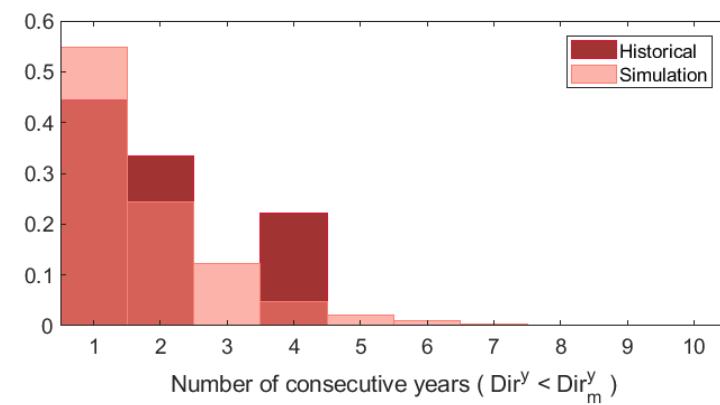
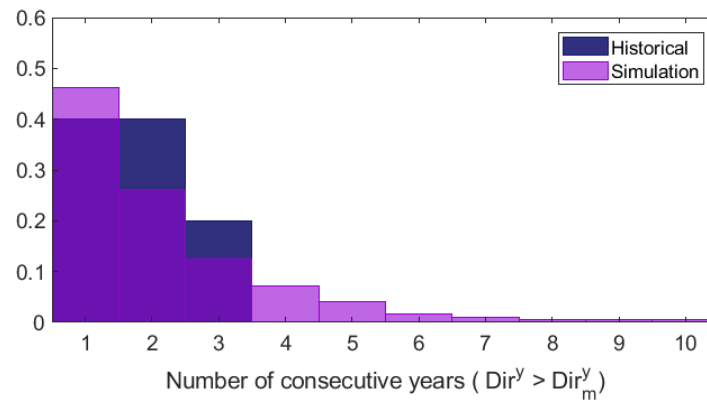
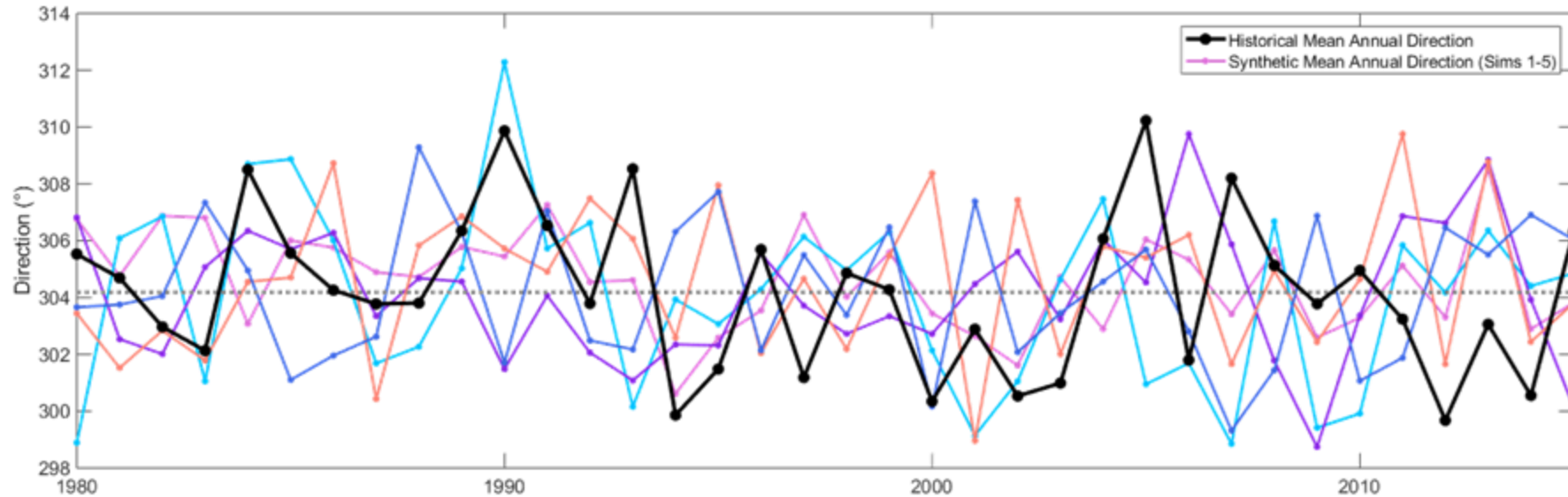


Applications

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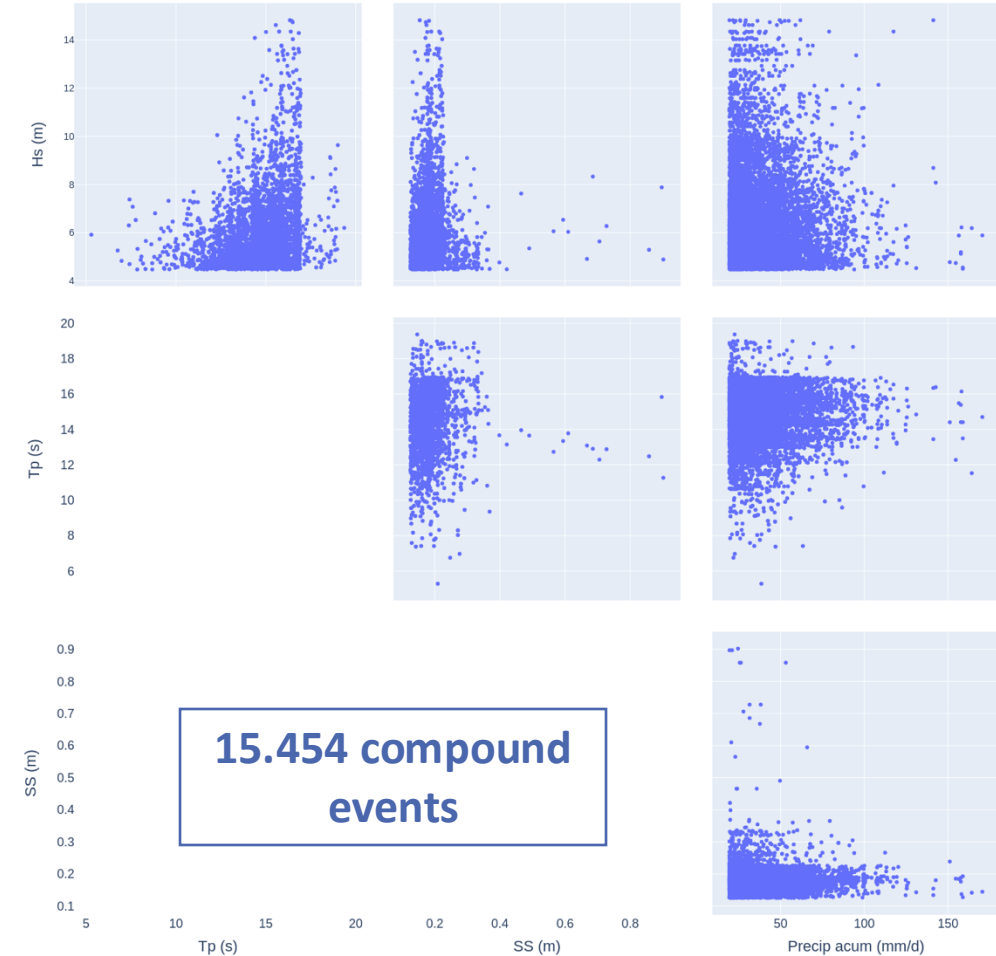
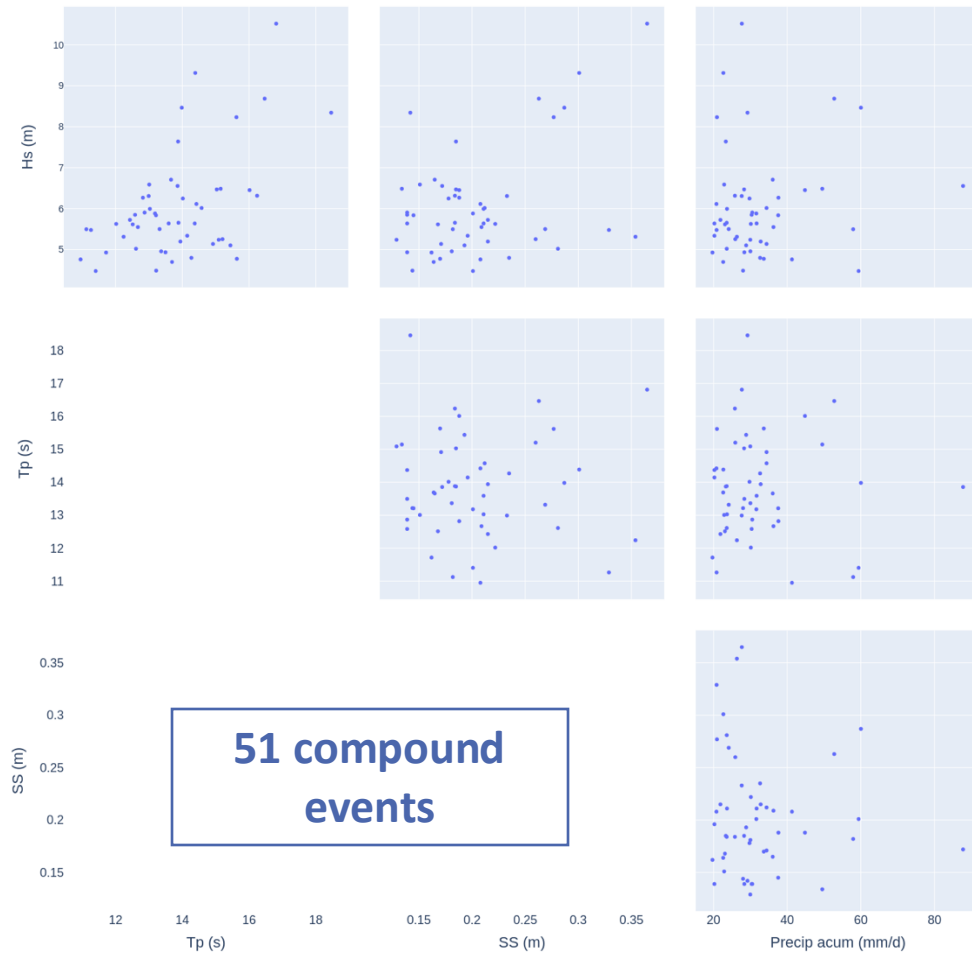
Annual mean wave direction
proxy



Applications

NUMBER OF COMPOUND EVENTS

Threshold = percentile 95 (Hs, SS, acum_precip)



Thank you

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<https://geocean.sci.unican.es/hyweb/>



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